Homework 1B: Practice exercises on class project (100 points)

Please do the exercises below for your system of choice (the one for which you are doing the team class project). **Please indicate clearly the members of your team, their Cornell NetID, and Github ID.**

No.	**First Name**	**Last Name**	**Cornell NetID**	**GithubID**
1	Janhavi	Gaikwad	jg2569	janhaviprof-bot
2	Anurag	Yadav	ay468	anuragyadav21
3	Kris	Singh	as4499	realAryavardhanSingh
4				
5				

Table 0: Project Team Members

A Family Of Next-Generation General Aviation Aircraft

Exercise 1: Project Proposal/Description (Anurag Yadav)

A. **Scope and Context**: *Define the scope and context of the project, including who is the organization behind this project.*

This project focuses on designing an architecture for next-generation family of general aviation (GA) aircraft that prioritizes resilience/reliability, safety, and lower carbon impact rather than just incremental performance gains. The scope is at the conceptual and system architecture level. Throughout this project we would not be dealing with engineering aspects like detailed aerodynamics or structural sizing, but instead defining the overall configuration, propulsion architecture, and safety logic of the system. The project is assumed to be led by a government-backed national transportation innovation body working in partnership with a mid-sized aerospace manufacturer with FAA Part 23(governs airworthiness standards) certification experience. The government's motivation is improving aviation safety margins and reducing emissions in regional aviation, while the industry partner ensures the concept remains realistic and certifiable.

B. **Proposed Topic**: *Provide a general overview of the system your team intends to design, appropriate to non-experts.*

Our team intends to design a modular family of 4–8 seat general aviation aircraft that is built around a resilience-first philosophy. Instead of relying on a single propulsion unit or traditional control architectures, the system will explore different propulsion systems, improved redundancy, and advanced flight envelope protection to reduce the probability and severity of catastrophic failures. The aircraft would be adaptable across different missions (trainer, personal use, light utility) using a shared platform. The goal is not to come up with the whole new propulsion system like nuclear or supersonic engines, but to rethink how the overall aircraft system is structured to make it safer, more efficient, and more sustainable.

C. Value Added: *Why is the system necessary or desirable? Be sure to reference a known problem/market need/anticipated need. What benefits do you expect from such a system, and how will you measure success?*

General aviation has improved over decades, but many aircraft still rely on architectures that are fundamentally similar to those designed 40–50 years ago. Loss-of-control accidents and single-engine failure risks still dominate fatal accident statistics. At the same time, carbon reduction pressures and operating cost challenges are increasing. This creates a need for aircraft that are not just more efficient, but structurally more resilient to worst-case scenarios.

The value of this system lies in reducing single-point failure risk, improving survivability during emergencies, and lowering lifecycle carbon impact through hybrid-electric integration. Success would be measured through metrics such as reduction in modeled fatal accident probability (based on historical data), improved propulsion redundancy index, fuel burn reduction compared to current GA aircraft in the same class, and lifecycle emission estimates. If the architecture demonstrates measurable safety margin improvement without unacceptable weight or cost penalties, it would represent meaningful progress beyond incremental upgrades.

D. **Current Reference Architecture:** *Describe some similar existing systems. These can be current iterations of the system you'll design, or can be other systems that exhibit functions/behaviors that your team intends to leverage in a novel domain. You can start thinking about whether there is a current reference architecture, what different*

The primary reference architecture for the project is the conventional single-engine piston aircraft such as the Cessna 172 or similar Part 23 certified aircraft. These systems rely on a single internal combustion engine, mechanical control surfaces (rudder, ailerons, flaps), and traditional aerodynamic stability principles. While highly mature and reliable, they inherently contain single-point propulsion failure risk. Safety improvements have largely come from avionics and pilot training rather than architectural redesign. Figures of merit in this architecture include cost, range, fuel burn, payload, and stall characteristics.

A second reference architecture for the project is modern composite GA aircraft such as the Cirrus SR-series, which integrates ballistic parachute recovery systems and advanced avionics. These systems represent incremental architectural innovation by adding whole-aircraft recovery redundancy. They show how survivability can be improved without radically changing propulsion. However, propulsion redundancy still remains limited in single-engine configurations.

A third relevant architecture comes from emerging hybrid-electric and distributed propulsion demonstrators (e.g., NASA X-57 concept). These systems explore multiple small electric motors

distributed along the wing, improving redundancy and aerodynamic control authority. While not yet widely commercialized, they demonstrate an alternative architecture where propulsion and control are partially integrated. Key design decisions here involve trade-offs between weight, redundancy, energy density, and certification feasibility.

Our project draws from these architectures but aims to combine distributed hybrid propulsion with risk-informed safety prioritization and modular platform design, rather than optimizing only one dimension.

E. **Data & Permissions**: *Briefly describe the sorts of resources (i.e., data) your team will need, why you'll need them (e.g., to develop metrics), and, most importantly, how you plan to obtain them. If the resources you need aren't available to you, please describe what alternatives you can use, why they're sufficient, and their limitations. Please mention permission to share your results with the class (in case you are working with a company).*

-The project will require historical general aviation accident databases https://aviation-safety.net/ (e.g., NTSB reports) to identify and rank high-severity failure modes. These data will be used to develop probabilistic risk metrics and define architecture priorities. We will also need publicly available performance data from existing GA aircraft to establish baseline comparisons for fuel burn, range, payload, and redundancy characteristics. Certification guidelines from FAA Part 23 documentation will be used to ensure feasibility constraints are understood early in the design process.

All data sources are publicly available, including NTSB databases, FAA regulatory documents, NASA research publications, and manufacturer performance specifications. If proprietary industry data are unavailable, publicly documented aircraft specifications and research demonstrator results will be used as proxies. While this may limit precision, it is sufficient for conceptual trade-space exploration. There are no confidentiality restrictions, and results can be shared openly with the class.

Exercise 2: Stakeholder networks (Kris Singh)

A. Fill out the table below with a list of the major 10 or so stakeholders. For each stakeholder, provide a succinct description of who they are and what their needs are with respect to your system.

Stakeholder Name	Description	Major needs w.r.t. the system
1. Private Pilots	Individuals licensed to operate the aircraft for personal or non-commercial travel.	● Simple and intuitive cockpit interface ● "Forgiving" flight characteristics (handling) ● Enhanced safety features (reduced workload)
2. Aircraft Owners	Individuals or entities who purchase and hold legal title to the aircraft (may also be pilots).	● Low direct operating costs (fuel/maintenance) ● High resale value (asset depreciation) ● Insurability and affordable premiums ● Reasonable acquisition cost
3. Flight Schools	Organizations providing pilot training (primary market for 4-seat trainers).	● High durability ● low hourly operating costs (profit margins) ● "forgiving" flight characteristics.
4. Passengers / Users	Individuals traveling in the aircraft for transportation or leisure.	● High safety assurance (survivability) ● Comfort and ergonomics ● Affordability of travel ● Speed and convenience
5. Mid-sized Aerospace Manufacturer	The industrial partner responsible for production and sales.	● A certifiable design ● manufacturability ● competitive advantage over existing Cessna/Cirrus fleets.
6. FAA (Regulators)	The US government body governing airworthiness standards (Part 23).	● Compliance with safety regulations

		• proof of reliability for novel propulsion • failure mode analysis. • Reliability
7. Govt. Innovation Body (Sponsor)	The government entity funding/backing the project to improve national transport.	• Measurable reduction in aviation accident rates • lower carbon emissions • technological prestige.
8. Aviation Insurance Companies	Companies insuring the hulls and liability for operators.	• Statistical proof of reduced risk to lower premiums • data on safety systems (e.g., envelope protection).
9. Maintenance (MRO) Providers	Mechanics and shops responsible for repairing the aircraft.	• Accessibility of parts • ease of diagnostics for systems • training on next-generation composite repair and avionics.
10. Airport Authorities	Managers of regional and general aviation airports.	• Reduced noise pollution (community relations) • compatibility with existing fueling and energy infrastructure
11. Investors / Shareholders	Financial backers of the manufacturing partner.	• Return on Investment (ROI) • market share growth in the GA sector • minimized development risk.
12. Local Communities	People living near GA airports.	• Reduced noise levels • elimination of lead pollution (from traditional AvGas). • Lower carbon emission

Table 1: List of stakeholders and needs

B. Organize this information in a stakeholder value network showing how value flows through exchanges between stakeholders.

Stakeholder Value Network: Flows and Exchanges

Central Node:

- **The Project / System Architecture:** A resilient, next-generation General Aviation (GA) aircraft family.

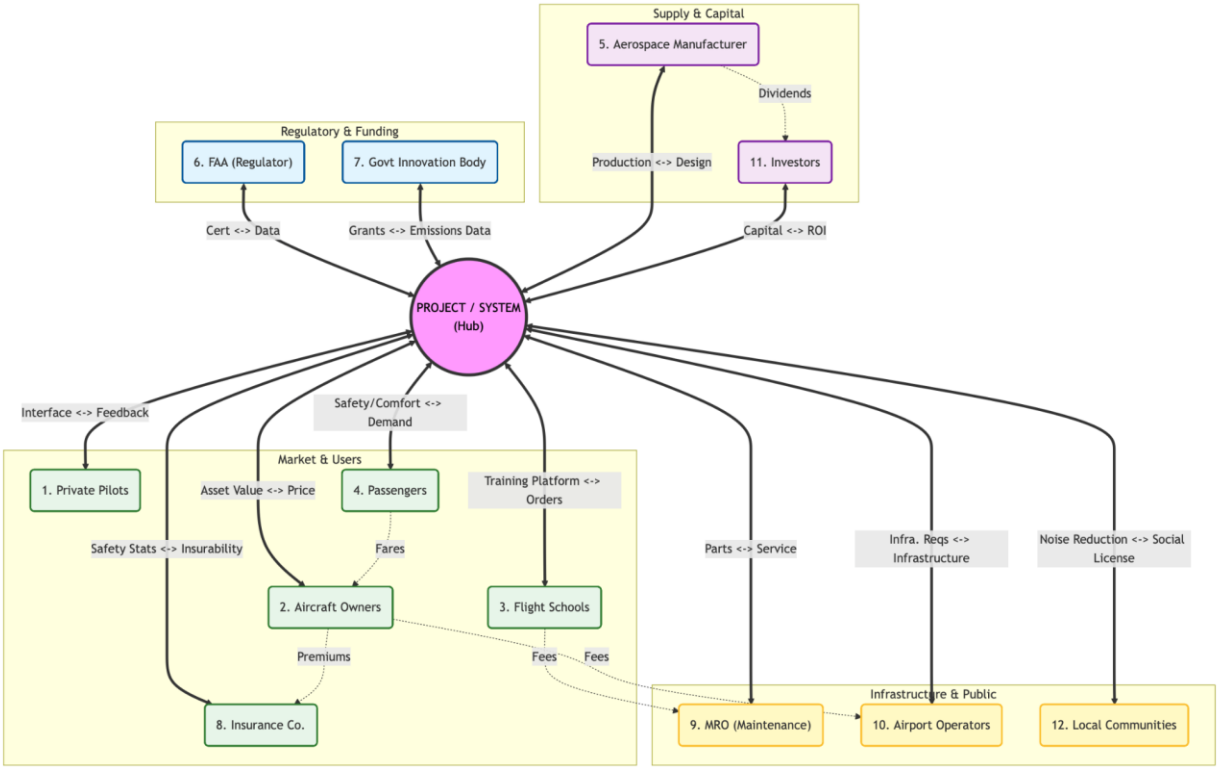
Radial Flows (Between System & Stakeholders):

- **1. Private Pilots**
  - **Flow to System:** Operational Feedback, User Experience Data.
  - **Flow from System:** Intuitive Interface, "Forgiving" Flight Characteristics (Handling).
- **2. Aircraft Owners**
  - **Flow to System:** Purchase Price (Revenue), Fleet Orders.
  - **Flow from System:** Low Operating Costs, High Resale Value, Resilient Asset.
- **3. Flight Schools**
  - **Flow to System:** Bulk Orders, Student Training Data.
  - **Flow from System:** Durable Training Platform, Safety Margins for Students.
- **4. Passengers / Users**
  - **Flow to System:** Market Demand, Public Confidence.
  - **Flow from System:** Safety Assurance (Survivability), Comfort (Low Noise/Vibration).
- **5. Aerospace Manufacturer (Industrial Partner)**
  - **Flow to System:** Production Capabilities, Existing Supply Chain.
  - **Flow from System:** Certifiable Design Specs, Competitive Market Advantage.
- **6. FAA (Regulators)**
  - **Flow to System:** Type Certification (Part 23 Airworthiness).
  - **Flow from System:** Compliance Evidence, Failure Mode Analysis Data.
- **7. Government Innovation Body (Sponsor)**
  - **Flow to System:** R&D Funding/Grants, Strategic Support.
  - **Flow from System:** Reduced Aviation Emissions Data, Improved National Safety Stats.
- **8. Insurance Companies**
  - **Flow to System:** Insurability Ratings, Risk Assessment Models.
  - **Flow from System:** Validated Safety Statistics (Envelope Protection Data).

- **9. MRO Providers (Maintenance)**
    - **Flow to System:** Service Network Availability, Maintenance Data.
    - **Flow from System:** Spare Parts, Repair Manuals (for New Tech/Avionics).
- **10. Airport Authorities**
    - **Flow to System:** Infrastructure Support (Hangar/Energy Access).
    - **Flow from System:** Reduced Noise Footprint, Standardization Compliance.
- **11. Investors**
    - **Flow to System:** Capital Investment.
    - **Flow from System:** Return on Investment (ROI), Intellectual Property (IP).
- **12. Local Communities**
    - **Flow to System:** Social License to Operate (Public Acceptance).
    - **Flow from System:** Reduced Environmental Impact (Noise/Lead).

<u>Rim Flows (Direct Exchanges Between Stakeholders):</u>

- **Owners ↔ Insurers:** Owners pay Premiums; Insurers provide Hull/Liability Coverage.
- **Owners ↔ Airports:** Owners pay Landing/Hangar Fees; Airports provide Runway Access.
- **Schools ↔ MROs:** Schools pay Maintenance Fees; MROs provide Repairs/Inspections.
- **Manufacturer ↔ Investors:** Manufacturer pays Dividends; Investors provide Equity.
- **Passengers ↔ Owners:** Passengers pay Fares/Cost Sharing; Owners provide Transportation.



C. (4 points) Comment on the structure of the network.

a. What is the network topology (e.g. hub-spoke, meshed)? What does that mean?

The topology is a **Hub-and-Spoke with Rim Connections** (also known as a Partial Mesh).

- **Hub:** The "Project/System" acts as the central hub, mediating the primary value exchange.
- **Spokes:** There are direct, bidirectional radial flows between the Project and every single stakeholder (e.g., FAA, Investors, Owners).
- **Rim Connections:** Unlike a pure hub-and-spoke, there are lateral connections between stakeholders (e.g., *Owners* ↔ *Insurers*, *Passengers* ↔ *Owners*).

b. Are there any features in the network such as hubs or loops? What do those mean?

Hubs: The **Project/System** is the primary hub. The **Pilots/Schools** (Customers) act as a secondary "Demand Hub," attracting connections from Insurers, MROs, and Airports.

Loops: The network is dense with loops. A "Value Loop" is any path that leaves the center and returns.

- *Simple Loop:* System → Owners → System.
- *Complex Loop:* System → Passengers → Owners → System.

D. For each value flow, assign a semi-quantitative (3-level) supply importance score and semi-quantitative (3-level) benefit score and combine them into a single number as seen in the lecture. Compute all value flow scores.

Scoring Key:

- **Importance (I):** 1 (Low), 2 (Medium), 3 (High/Critical).
- **Benefit (B):** 1 (Low), 2 (Medium), 3 (High/Transformative).
- **Score:** I×B.

Flow Source → Target	**Value Object**	**Imp (I)**	**Ben (B)**	**Score**	**Justification**
System → Owners	**Resilient Asset**	3	3	**9**	Owners need a durable asset with high resale value to justify the investment.
Owners → System	**Purchase Price**	3	3	**9**	Primary revenue source. Essential for financial viability.
System → Pilots	**Intuitive Interface**	3	3	**9**	Critical for safety; prevents loss-of-control accidents.
Pilots → System	**Feedback**	2	2	**4**	Validates design choices but is not a direct financial

					transaction.
System → Schools	**Training Platform**	3	3	**9**	The core product for flight schools; determines their fleet uptime.
Schools → System	**Fleet Orders**	3	3	**9**	Bulk sales are a major revenue driver.
System → Pax	**Safety Assurance**	3	3	**9**	Passengers will not fly if the system is perceived as unsafe.
Pax → System	**Market Demand**	3	3	**9**	Public confidence drives the entire market need for the aircraft.
System → FAA	**Compliance Data**	3	2	**6**	Mandatory for legal operation. Benefit is "permission."
FAA → System	**Type Certification**	3	3	**9**	The absolute "Gateway." No certification = No business.
System → Govt	**Emissions Data**	2	2	**4**	Validates the govt's investment.

Govt → System	**R&D Grants**	2	3	**6**	Critical early-stage funding that lowers risk.
System → Insurers	**Safety Statistics**	2	2	**4**	Necessary to prove "Resilience" and lower rates.
Insurers → System	**Insurability Rating**	2	2	**4**	Market enabler; allows customers to afford the aircraft.
System → MRO	**Parts & Manuals**	3	2	**6**	Essential for after-market support.
MRO → System	**Service Network**	2	2	**4**	Availability of repair shops makes the aircraft attractive to buy.
System → Airports	**Infra. Reqs**	1	2	**2**	Ensuring compatibility with fuel/energy standards.
Airports → System	**Infrastructure**	2	2	**4**	Hangar space and energy access are prerequisites for use.
System → Investors	**ROI & IP**	3	3	**9**	Investors need a return or they pull

					funding.
Investors → System	**Project Capital**	3	3	**9**	Direct funding for the design/architecture phase.
Investors → Mfg	**Equity Investment**	3	3	**9**	Capital for the manufacturer to build factories/tooling.
Mfg → Investors	**Dividends**	2	3	**6**	Financial return on the manufacturing partner's side.
System → Comm.	**Noise Reduction**	1	2	**2**	Good for PR, helps with airport curfews.
Comm. → System	**Social License**	1	1	**1**	Passive acceptance.
Mfg → System	**Production Cap.**	3	3	**9**	The partner builds the actual plane. Critical dependency.
System → Mfg	**Design Specs**	3	2	**6**	The "blueprint" the team provides to the manufacturer.
Owners → Insurers	**Premiums**	3	2	**6**	Mandatory cost for owners.

Insurers → Owners	**Coverage**	3	3	**9**	Owners cannot legally/financially fly without it.
Owners → Airports	**Fees**	2	1	**2**	Standard transaction.
Airports → Owners	**Runway Access**	3	3	**9**	No runway = No flying.
Schools → MRO	**Maint. Fees**	3	2	**6**	Major operating cost for schools.
MRO → Schools	**Repairs**	3	3	**9**	Schools are grounded (lose money) if repairs aren't done.
Pax → Owners	**Fares**	3	3	**9**	The revenue model that justifies the owner buying the plane.
Owners → Pax	**Transportation**	3	3	**9**	The service being provided.

Table 2: Value Flow score

E. **(4 points)** Find **all value loops (cycles_ and their corresponding scores** by any method of your choice (explain which: inspection, matrix multiplication, brute-force). What are the most important value loops? Discuss.

Method Used: Computational Brute-Force (Python/NetworkX) To ensure exhaustiveness, we used a computational approach (Python with the NetworkX library) to identify all elementary circuits (simple cycles) in the directed graph. We then calculated the "Loop Score" by summing the specific Importance × Benefit weights of all edges contained within each cycle. This method avoids the bias of visual inspection and uncovers complex multi-stakeholder dependencies.

Top Identified Value Loops:

Based on the computational ranking, the three most critical loops are:

1. The "Passenger Confidence" Loop (Rank 1, Score: 31)

- **Path:** Owners → Pax → System → Insurers → Owners
- **Flows:**
    1. Owners provide *Transportation* to Passengers (Score: 9)
    2. Passengers provide *Market Demand* to System (Score: 9)
    3. System provides *Safety Stats* to Insurers (Score: 4)
    4. Insurers provide *Coverage* to Owners (Score: 9)
- **Discussion:** This 4-node loop achieved the highest score, illustrating that the entire ecosystem is driven by the end-user (Passengers) but gated by risk (Insurers). It links the **Market** (Passenger demand) with the **Financial**(Insurance coverage) and the **Technical** (Safety stats). It proves that even if the Owners *want* to buy the plane, they cannot operate it for Passengers unless the System proves its safety to Insurers first.

2. The "Capital Expansion" Loop (Rank 4, Score: 27)

- **Path:** Manufacturer → System → Investors → Manufacturer
- **Flows:**
    1. Manufacturer provides *Production Capacity* to System (Score: 9)
    2. System provides *ROI/IP* to Investors (Score: 9)
    3. Investors provide *Equity Investment* to Manufacturer (Score: 9)
- **Discussion:** This loop (tied for 4th) highlights the high capital requirements of the project. It demonstrates that the "System" (Project Team) acts as the economic vehicle that converts Manufacturing capability into Investor ROI. Without this loop, the physical aircraft cannot be built, regardless of how much passenger demand exists.

3. The "Operational Sustainment" Loop (Rank 7, Score: 24)

- **Path:** Schools → System → MRO → Schools
- **Flows:**
    1. Schools provide *Fleet Orders* to System (Score: 9)
    2. System provides *Parts & Manuals* to MRO (Score: 6)
    3. MRO provides *Repairs* to Schools (Score: 9)
- **Discussion:** This loop represents the **Service Ecosystem**. For a high-utilization operator like a Flight School, the aircraft is a liability if it cannot be repaired. This loop ranks highly because it captures the lifecycle dependency: if the System team does not provide valid

technical data to MROs, the MROs cannot service the Schools, and the Schools will stop ordering the aircraft.

Overall Discussion of Network Structure: The analysis reveals that the network is dominated by **complex (3+ node) cycles** rather than simple bilateral trades. The top loops all involve intermediaries—Passengers driving demand for Owners, or MROs enabling operations for Schools. This implies the architecture cannot succeed as a standalone product; it requires a synchronized ecosystem where the "System" facilitates value for third parties (Insurers, Passengers, MROs) to ensure the primary customers (Owners/Schools) remain viable.

F. **(4 points)** Identify the **most important stakeholders** and briefly describe the method used for that (e.g., weighted stakeholder occurrence). Discuss, including the limitations of this method.

Method Used: Weighted Stakeholder Occurrence To identify the critical stakeholders, we utilized the **Weighted Stakeholder Occurrence** method. This was implemented computationally using Python to calculate the "Weighted Degree" for each node in the network.

- **Process:** We summed the total value scores (Importance×Benefit) of all value flows entering and leaving each stakeholder.
- **Rationale:** This method quantifies a stakeholder's influence by measuring not just the *number* of connections they have, but the cumulative *value intensity* of those transactions.

Results: Most Important Stakeholders

Based on the computational analysis, the hierarchy of importance is:

1. **Aircraft Owners (Score: 62) – The Financial Nexus**
    - **Rank:** 1st (Dominant)
    - **Context:** This result highlights a critical distinction in the system architecture: the *User* (Pilot) is not necessarily the *Customer* (Owner). Owners achieved the highest score because they act as the "Financial Hub" of the network. They connect the **Supply Side** (paying the System for aircraft), the **infrastructure Side** (paying Airports/Insurers), and the **Demand Side** (collecting fares from Passengers). If the Owner cannot make a business case, the entire network dissolves.
2. **Passengers (Score: 36) – The Value Drivers**
    - **Rank:** 2nd

- **Context:** Passengers rank highly because they are the ultimate source of revenue for the Owners and the ultimate source of demand for the System. Their high score reflects that safety and comfort are not just "nice-to-haves" but are critical drivers of market viability.

3. **Flight Schools & Investors (Score: 33) – The Enablers**
    - **Rank:** Tied for 3rd
    - **Context:**
        - **Flight Schools:** Similar to Owners, they act as fleet operators who bridge the gap between the Manufacturer (System) and the MRO/Student ecosystem.
        - **Investors:** Their high score reflects the capital-intensive nature of the project. While they have fewer connections, the high *intensity* of the "Capital" and "ROI" flows keeps them near the top.

Discussion & Limitations of this Method

The results expose two significant limitations of the **Weighted Stakeholder Occurrence** method:

- **The "Operator vs. Check-Writer" Bias (The Pilot Drop):**
    - **Observation: Private Pilots** ranked **Rank 9** (Score 13), far below Owners and Passengers.
    - **Analysis:** This method devalues stakeholders who interact with the system purely operationally rather than financially. While the Pilot is critical for *safety* (they fly the plane), they do not generate or transfer capital in this model. The method risks under-prioritizing Human Factors (pilot interface) because the "money" flows elsewhere.
- **The "Gatekeeper" Paradox (FAA Ranking):**
    - **Observation:** The **FAA (Regulator)** ranked **8th** (Score 15), despite being a regulatory absolute.
    - **Analysis:** The method assumes that "More Value Flow = More Importance." It treats influence as *additive*. However, the FAA is a "Boolean" stakeholder. They have only one major loop (Certification). If that single loop fails, the project ends immediately. The method fails to capture the **criticality of constraints** versus the **volume of transactions**.

Exercise 3: Needs to Goals (Janhavi Gaikwad)

A. Write a hierarchy of the 15-20 or so most important goals for your system.

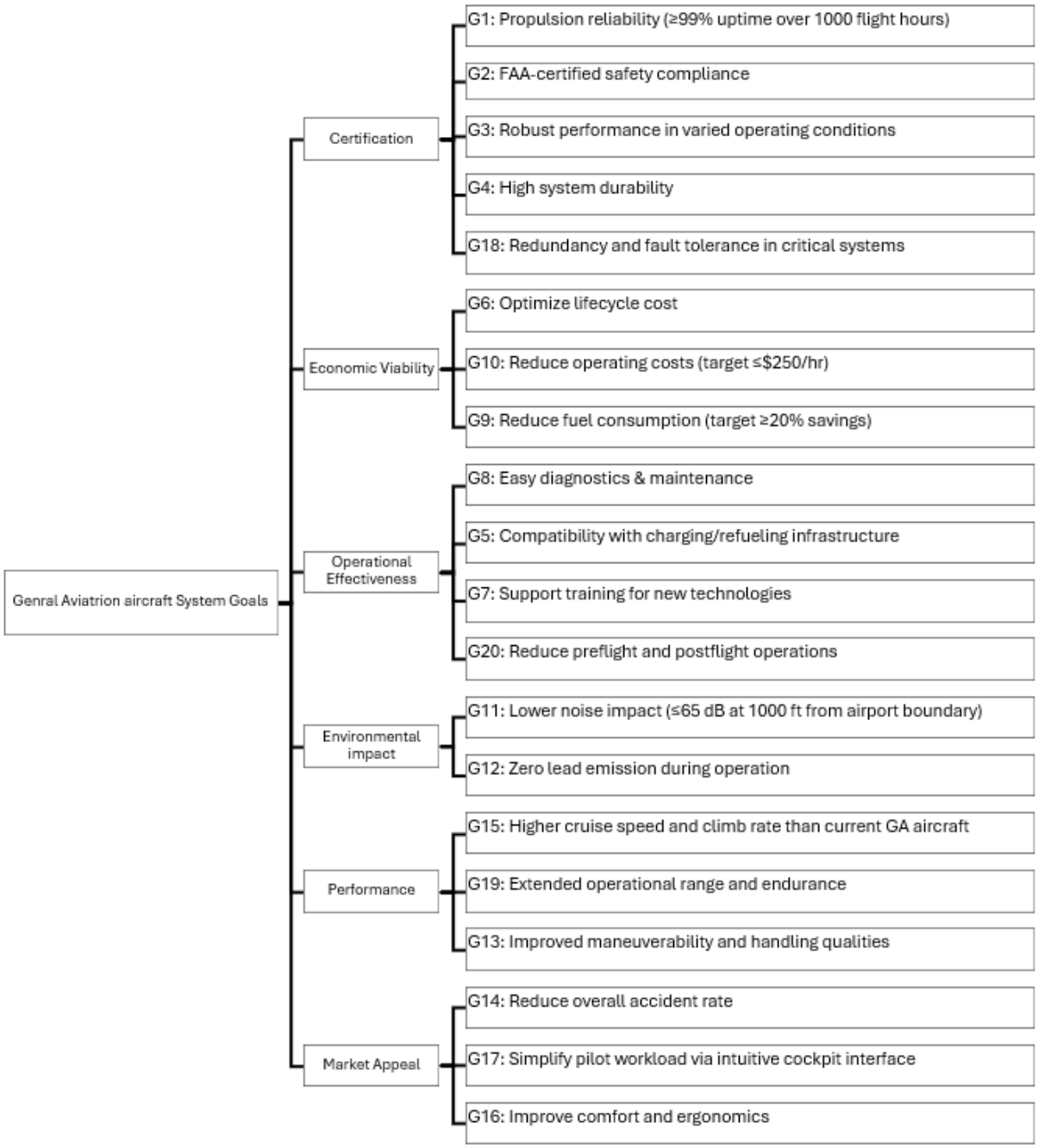
Goal Id	Goal Statement	Traceability to needs
G1	Must ensure propulsion reliability (≥99% operational uptime over 1000 flight hours)	• Proof of reliability for propulsion • Enhanced safety
G2	Must achieve FAA-certified safety compliance	• Compliance with safety regulations • A certifiable design • Data on safety systems
G3	Must provide robust performance in varied operating conditions	• Enhanced safety • Measurable reduction in aviation accident rates • Compliance with safety regulations • A certifiable design • High safety assurance (survivability)
G4	Must provide high system durability	• High durability • Reliability • High resale value (asset depreciation)
G5	Must maintain compatibility with charging/refueling infrastructure	• Compatibility with existing or simple charging infrastructure
G6	Must optimize the overall lifecycle cost	• Manufacturability • Minimized development risk • Reasonable acquisition cost
G7	Must support training for new technologies	• Training on new high-voltage architectures

G8	Should enable easy diagnostics & maintenance	• Ease of diagnostics for systems • Accessibility of parts • Lower maintenance costs
G9	Should reduce fuel consumption (target ≥20% savings against conventional piston aircraft)	• Lower fuel costs • Lower carbon emissions
G10	Should reduce operating costs (target ≤ $250/hr)	• Low hourly operating costs • Return on Investment (ROI) • Low direct operating costs (fuel/maintenance) • Affordability of travel
G11	Should lower noise impact (≤65 dB at 1000 ft from airport boundary)	• Reduced noise pollution (community relations) • Reduced noise levels
G12	Should ensure lower carbon & lead emission during aircraft operation	• Elimination of lead pollution (from traditional AvGas) • Lower carbon emissions
G13	Should improve maneuverability and handling qualities	• “Forgiving” flight characteristics
G14	Should reduce overall accident rate	• Measurable reduction in aviation accident rates • Statistical proof of reduced risk to lower premiums
G15	Should achieve higher cruise speed and climb rate than current GA aircraft.	• Market share growth in the GA sector • Competitive advantage over existing Cessna / Cirrus Aircraft fleets • Speed and convenience
G16	Might improve comfort and ergonomics of the air craft	• Market share growth in the GA sector • Competitive advantage over existing Cessna / Cirrus Aircraft fleets. • Comfort and ergonomics

G17	Might simplify pilot workload via intuitive cockpit interface	• Simple and intuitive cockpit interface
G18	Must ensure redundancy and fault tolerance in critical system	• Enhanced safety
G19	Should extend operational range and endurance.	• Market share growth in the GA sector • Competitive advantage over existing Cessna / Cirrus Aircraft fleets
G20	Should reduce preflight and postflight operations	• Competitive advantage over existing Cessna / Cirrus Aircraft fleets • Low hourly operating cost • Affordability of travel

Table 3: Hierarchy of system goals

B. Visualize your table as a hierarchical tree.



C. Discuss how your list of goals addresses the criteria for good sets of goals seen in class.

I. Solution Neutral : The goals are written without prescribing a solution.

Ii. Target values : Goals are provided with target values and metrics wherever applicable.

Iii. Traceable : All goals are traceable to stakeholder needs.

Iv. Consistent : Relative priorities of goals are mentioned in the table

V. Complete : If all goals are met then all prioritized stakeholder needs will be met.

Exercise 4: Concept (Janhavi Gaikwad)

A. Generate alternative concepts for your system using the creativity techniques seen in class and summarize the information in <u>the table below</u>.

a. For each concept, provide a short description as well its pros and cons.

Concept Name	Description	Pros	Cons
Modern Performance GA	Low-wing composite airframe with single turboprop; emphasizes proven high-performance design, speed, climb, and efficiency. Combines conventional modern GA and high-performance concepts.	• Proven and certifiable design • Higher cruise speed and climb rate • Moderate development risk	• Limited redundancy • Not optimized for environmental sustainability
Resilient Hybrid GA	Hybrid-electric propulsion with redundancy, modular airframe, and extreme safety features. Merges Eco GA, Resilient Green, Ultra-Long-Range Hybrid, and Extreme Safety concepts. Focused on green, safe, and resilient design.	• High safety and redundancy • Lower carbon emissions • Flexible for multi-mission use • Extended endurance	• Complex powertrain • Certification challenges • Higher development and acquisition cost
Twin Engine Safety GA	High- or low-wing GA with twin piston or turboprop engines; combines Utility GA and Twin Engine Redundancy concepts. Prioritizes propulsion redundancy and safety.	• Excellent safety through propulsion redundancy• Reliable and familiar for operators• Simple certification path	• Higher fuel burn • Heavier airframe • Less environmentally friendly
Modular Multi-Mission GA	Configurable airframe with automated and redundant safety systems; merges Modular Aircraft, AI Assisted Safety, Redundant Control	• Highly flexible for multiple missions (trainer, utility, personal) • Scalable production	• Moderate design complexity • May sacrifice aerodynamic efficiency

	Systems, and Automation Assisted Safety concepts. Focused on flexibility, modularity, and smart systems.	• Integrated advanced safety systems	• Higher development effort for automation and certification
Innovative Airframe GA	Radical airframe layouts (blended wing, canard) with fully electric or low-cost configurations; merges Novel Airframe, Short-Range Electric, Low-Cost Minimalist, Green, and Budget Electric Trainer concepts. Focused on innovation and market differentiation.	• High innovation potential • Potential aerodynamic efficiency gains • Distinctive market identity • Environmentally friendly (electric / low-carbon)	• High development risk • Unproven certification path • Complex pilot training • May have limited payload or range

Table 4: List of alternative concepts

b. Explain how you came up with these concepts.

To generate alternative concepts for our GA aircraft we used multiple creativity techniques as follows

1. <u>Abstraction, Composition & Recursion:</u>

   Airframe and propulsion are critical decisions at the early stage of design. Hence we started by creating different alternative concepts for the airframe and propulsion and combined feasible option suitable for our use case(small passenger aircraft)

Concept	**Airframe**	**Propulsion**
C1: Classic GA	Low-wing conventional	Single piston
C2: Utility GA	High-wing conventional	Twin piston
C3: Modern GA	Low-wing composite	Single turboprop
C4: Eco GA	Low- or mid-wing composite/modular	Hybrid-electric
C5: Short-Range Electric GA	Mid-wing blended	Fully electric

Table 5: Abstraction Table for concepts

2. Analogy

   For generating next set of concepts we drew inspiration from other industries:
   AI driver assist system in cars : Automation for safety
   Automation redundancy : Multi redundant control systems
   Multi mission vehicles : Flexible aircraft roles
   Based on these analogies we generate concepts including
   C6 : AI assisted safety aircraft
   C7 : Redundancy control system concept
   C8 : Automation assisted safety aircraft

3. Morphology Matrix

   Then we identified key design dimensions loosely tied to goals and combined feasible options to explore new concepts:

Dimension	**Options**
Propulsion	Single / Twin / Distributed
Safety	Basic / Redundant / Automated
Mission	Single-role / Modular
Cost	Low-cost / Premium

Table 6: Morphology Matrix

   Concepts derived from the morphological matrix:
   C9 : Twin Engine Redundancy Aircraft
   C10 : Low-Cost Minimalist Aircraft
   C11 : Modular Aircraft

4. Extremes and Inverses

   We explored extreme scenarios to generate innovative concepts:
   - Safety vs. Cost
   - Performance vs. Environmental Impact

   Resulting concepts:
   C12 : Extreme Safety Aircraft
   C13: High-Performance Aircraft
   C14: Green Aircraft

5. <u>Brainstorming</u>

Finally, we reviewed all previously generated concepts and identified potential gaps or missing ideas. Using a team brainstorming session, we generated concepts that combine features or push design boundaries:

C15 : Resilient Green GA – combines hybrid-electric propulsion, redundancy, and modularity

C16: Novel Airframe GA – explores non-traditional airframe layouts (e.g., blended-wing, canard)

C17: Ultra-Long-Range Hybrid GA – hybrid-electric optimized for extended endurance

C18: Budget Electric Trainer – affordable electric GA aircraft for flight schools

Now these 18 concepts can be combined and clustered to form top 5 concept:

Final Concept Name	**Merged Original Concepts**	**Description / Reason for Merging**
FC1 : Modern Performance GA	C3: Modern GA, C13: High-Performance Aircraft	Focused on conventional yet high-performing airframe and propulsion (composite, turboprop), emphasizing speed, climb, and efficiency. Merged performance-oriented concepts.
FC2 : Resilient Hybrid GA	C4: Eco GA, C15: Resilient Green GA, C17: Ultra-Long-Range Hybrid GA, C12: Extreme Safety Aircraft	Combines hybrid-electric powertrain, redundancy, modularity, and extreme safety features. Represents “green + resilient” cluster; balances safety, environmental goals, and multi-mission endurance.
FC3 : Twin Engine Safety GA	C2: Utility GA, C9: Twin Engine Redundancy Aircraft	Merged twin-engine redundancy concepts; prioritizes safety via propulsion redundancy for short-to-medium missions.
FC4 : Modular Multi-Mission GA	C11: Modular Aircraft, C6: AI Assisted Safety Aircraft, C8: Automation Assisted Safety Aircraft, C7: Redundant Control Systems Concept	Combines modularity and automation for flexibility in roles (trainer, utility, personal) while integrating smart safety features and redundant systems.

FC5 :Innovative Airframe GA	C16: Novel Airframe GA, C5: Short-Range Electric GA, C10: Low-Cost Minimalist Aircraft, C14: Green Aircraft, C18: Budget Electric Trainer	Merges radical innovation, alternative airframes, fully electric or low-cost designs, and extreme environmental focus. Captures highly innovative, disruptive concepts.

Table 7: Final Concept Table

B. (**12.5 points**) Select a concept using **one** of the concept selection tools seen in class. You may use the list of goals from the previous exercises as criteria.

a. Please justify your decision for the final selection. For example, if you use a Pugh matrix, include the table with all scores.Discuss the limitations of the method you use

After evaluating the five final concepts using a weighted Pugh matrix based on 20 system goals, FC2: Resilient Hybrid GA was selected as the preferred concept because of the following reasons:

- Highest Weighted Score: FC2 achieved the top total score (+43), excelling in critical "Must" and "Should" goals such as propulsion reliability, FAA safety compliance, redundancy, accident rate reduction, and environmental impact.

- Balanced Performance: Unlike FC1, which emphasizes speed, or FC5, which focuses on radical innovation, FC2 balances safety, environmental sustainability, and operational flexibility, making it practical for small passenger aircraft.
- Future-Proof Design: It integrates hybrid-electric propulsion, modularity, and extreme safety features, aligning with emerging regulations, low-carbon aviation trends, and multi-mission requirements.
- Stakeholder Alignment: FC2 addresses top-priority needs of pilots, flight schools, regulators, and innovation bodies without major trade-offs in cost or feasibility

FC2: Resilient Hybrid GA offers a safe, sustainable, and flexible aircraft architecture, addressing both current and anticipated general aviation requirements.

Goal #	**Goal Statement**	**Weight**	**FC1 (Base)**	**FC2**	**FC3**	**FC4**	**FC5**
G1	Must ensure propulsion reliability (≥99% operational uptime over 1000 flight hours)	3	0	+2	+2	+1	–1

G2	Must achieve FAA-certified safety compliance	3	0	+2	+1	+1	0
G3	Must provide robust performance in varied operating conditions	3	0	+1	+1	+1	–1
G4	Must provide high system durability	3	0	+2	+2	+1	–1
G5	Must maintain compatibility with charging/refueling infrastructure	3	0	+1	0	+1	+1
G6	Must optimize the overall lifecycle cost	3	0	–1	0	–1	+1
G7	Must support training for new technologies	3	0	+1	0	+2	+1
G8	Should enable easy diagnostics & maintenance	2	0	+1	+1	+2	0
G9	Should reduce fuel consumption (target ≥20% savings)	2	0	+2	0	+1	+2
G10	Should reduce operating costs (≤ $250/hr)	2	0	0	0	–1	+2
G11	Should lower noise impact (≤65 dB)	2	0	+1	0	+1	+2
G12	Should ensure zero lead emission	2	0	+1	0	+1	+2
G13	Should improve maneuverability and handling qualities	2	0	+1	+2	+1	0
G14	Should reduce overall accident rate	2	0	+2	+1	+1	–1
G15	Should achieve higher cruise speed and climb rate	2	0	+1	+2	0	–1
G16	Might improve comfort and ergonomics	1	0	+1	+1	+1	0
G17	Might simplify pilot workload via intuitive cockpit	1	0	+1	0	+2	+1
G18	Must ensure redundancy and fault tolerance in critical system	3	0	+2	+2	+1	0
G19	Should extend operational range and endurance	2	0	+2	+1	+1	–1

G20	Should reduce preflight and postflight operations	2	0	+1	0	+2	0
	Weighted score		**0**	**43**	**23**	**30**	**20**

Table 8: Pugh Matrix

Limitation of Pugh matrix:

While pugh matrix provides a structured approach to concept comparison it has several limitations

<u>Subjectivity in Scoring:</u>
Assigning values from –2 to +2 for each criterion depends on the evaluators' judgment. Different teams might produce different scores for the same concepts.

<u>Dependence on Baseline Selection:</u>
The comparison is relative to the chosen baseline concept (FC1). Changing the baseline could alter the relative rankings.

<u>Weight Assignment Bias:</u>
Goal weights (Must = 3, Should = 2, Might = 1) heavily influence the final ranking. Small adjustments to weights could change which concept ranks highest.

<u>Limited Granularity:</u>
Using only five scoring levels (–2 to +2) may oversimplify subtle differences between concepts.

<u>Interdependency of Goals:</u>
Some goals are correlated (e.g., fuel efficiency and carbon emissions), but the Pugh matrix treats each goal independently, potentially overemphasizing certain benefits.